Midterm [75min]

1. If 1 + 1 = 2 or 1 + 1 = 3, then 2 + 2 = 3 and 2 + 2 = 4. **F**
2. Write the truth table for the proposition $\neg(r \to \neg q) \vee (p \wedge \neg r)$.

p	q	r	$\neg(r \to \neg q) \vee (p \wedge \neg r)$
T	T	T	**T**
T	T	F	**T**
T	F	T	**F**
T	F	F	**T**
F	T	T	**T**
F	T	F	**F**
F	F	T	**F**
F	F	F	**F**

3. Write a proposition with three variables p, q, and r that is true when p and r are true and q is false, and false otherwise.

$\boldsymbol{p \wedge \neg q \wedge r}$ / 다른 Proposition 을 작성했어도 Truth value 가 Equivalent 하다면 정답 처리됩니다.

(모범 답안이 아닌 다른 답안을 작성한 경우, Truth table 을 작성하여 조교 메일로 보내주세요.)

4. Write a proposition with three variables p, q, and r that is true when exactly one of the three variables is true, and false otherwise.

$\boldsymbol{(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r)}$ /

다른 Proposition 을 작성했어도 Truth value 가 Equivalent 하다면 정답 처리됩니다.

(모범 답안이 아닌 다른 답안을 작성한 경우, Truth table 을 작성하여 조교 메일로 보내주세요.)

5. Write a proposition with three variables p, q, and r that is never true. $\boldsymbol{(p \wedge \neg p) \vee (q \wedge \neg q) \vee (r \wedge \neg r)}$

다른 Proposition 을 작성했어도 Truth value 가 Equivalent 하다면 정답 처리됩니다.

(모범 답안이 아닌 다른 답안을 작성한 경우, Truth table 을 작성하여 조교 메일로 보내주세요.)

6. For propositions p,q,r, $p \to (q \to r)$ is equivalent to $(p \to q) \to r$. **F**

7. Write a proposition equivalent to $p \to q$ using only $p,q,\neg$ and the connective $\wedge$. $\boldsymbol{\neg(p \wedge \neg q)}$
8. For propositions p,q,r , $p \to (\neg q \wedge r)$, $\neg p \vee \neg(r \to q)$ are logically equivalent. **T**

9. $P(x,y)$ means "$x + 2y = xy$", where x and y are integers. $\neg\forall x\exists y\neg P(x,y)$ is False. **T**

10. Suppose $P(x,y)$ is a predicate and the universe for the variables x and y is {1,2,3}. Suppose $P(1,3)$, $P(2,1)$, $P(2,2)$, $P(2,3)$, $P(2,3)$, $P(3,1)$, $P(3,2)$ are true, and $P(x,y)$ is false otherwise. $\forall y\exists x(P(x,y) \to P(y,x))$ is true. **T**

11. An argument of the following form is not valid because p false and q true yield true hypotheses but a false conclusion. **T**

$$\begin{array}{c} p \to q \\ \neg p \\ \hline \therefore \neg q \end{array}$$

12. The following argument is valid: **F**

$$\begin{array}{c} p \to r \\ q \to r \\ q \vee \neg r \\ \hline \therefore \neg p \end{array}$$

13. The following argument is valid. **T**

- She is a Math Major or a Computer Science Major.
- If she does not know discrete math, she is not a Math Major.
- If she knows discrete math, she is smart.
- She is not a Computer Science Major.
- Therefore, she is smart.

14. The premises "Every student in this class passed the first exam" and "Alvina is a student in this class" imply the conclusion "Alvina passed the first exam." by Universal instantiation. **T**

15. Consider the following theorem: If n is an even integer, then $n + 1$ is odd. To show this, suppose $n = 2k$ but $n + 1 = 2l$. We then have that $2k + 1 = 2l$. Therefore, the theorem is true. This proof has been done by a proof by contradiction. **T**

16. The following f is a function from the set of all bit strings to the set of integers: if $f(S)$ is the largest integer i such that the ith bit of S is 0 and $f(S) = 1$ when S is the empty string (the string with no bits).

F. f not defined for the string of all 1's, for example $S = 11111$.

17.For any function $f : A \to B$, define a new function $g : P(A) \to P(B)$ as follows: for every $S \subseteq A$, $g(S) = \{f(x) \mid x \in S\}$. f is onto if and only if g is onto. **T**

18. The following algorithm takes a list of n integers ($n \geq 1$) and finds the average of the largest and smallest integers in the list. **T**

procedure $avgmaxmin(a_1,\ldots,a_n$: integers)
$max: = a_1$
$min: = a_1$
for $i := 2$ **to** n
begin
if $a_i > max$ **then** $max: = a_i$
if $a_i < min$ **then** $min: = a_i$
end
$avg: =(max + min)/2$.

19. Suppose that the only currency were 3-dollar bills and 10-dollar bills. The following proof to show that any dollar amount greater than 17 dollars could be made from a combination of these bills is correct.
Proof: $P(18)$: Eighteen dollars can be made using six 3-dollar bills. $P(k) \to P(k + 1)$: Suppose that k dollars can be formed, for some $k \geq 18$. If at least two 10-dollar bills are used, replace them by seven 3-dollar bills to form $k + 1$ dollars. Otherwise (that is, at most one 10-dollar bill is used), at least three 3-dollar bills are being used, and three of them can be replaced by one 10-dollar bill to form $k + 1$ dollars. **T**

20. Let $s_1,s_2\ldots,s_{101}$ be 101 bit strings of length at most 9. Then there exist two strings, s_i and s_j, where $i \neq j$, that contain the same number of 0s and the same number of 1s. (For example, strings 001001 and 101000 contain the same number of 0s and the same number of 1s.) As we know, there are ten possible lengths a bit string can have, that is, 0,1,2,...,9. Because there are 101 bit strings, there is a length number k such that at least 11 bit strings have length k. The number of 0s in these 11 bit strings must be one of the ten numbers 0,1,2,...,9. Therefore, there are at least two bit strings s_i and s_j, with the same number of 0s. Since s_i and s_j have the same length, k, they both have the same number of 1s. **T**

21. The following algorithm correctly computes 3^{2^n} where n is a nonnegative integer. **F**

procedure $power(n$: nonnegative integer)
if $n = 0$ **then** $power(n) := 3$
else $power(n) := power(n-1) \cdot power(n-1)$.

22. A recursive definition of the set $\{3,7,11,15,19,23,\ldots\}$ is XXX.

XXX = $3 \in S$; $x \in S \to x + 4 \in S$

23. A recursive definition of the set of strings 1,111,11111,1111111,....is XXX.

XXX = $1 \in S$; $x \in S \to x11 \in S$ (or $x \in S \to 100x + 11 \in S$).

24, A computer randomly prints three-digit codes, with no repeated digits in any code (for example, 387, 072, 760). The minimum number of codes that must be printed in order to guarantee that at least six of the codes are identical is XXX.

XXX = 3601